QX4001 – Molecules to Materials

Rotation 1 – Question Set 3

1) Draw the overlap of atomic orbitals leading to the formation of molecular orbitals in the following cases (draw the shapes of the atomic orbitals and all their respective molecular orbitals, with attention to orbital phase, position of nuclei and nodes):

   i) 1s + 2p
   ii) 2p + 2p (frontal overlap)
   iii) 2p + 2p (lateral overlap)

2) Draw a molecular orbital diagram for a Be2 molecule and comment on whether this molecule could exist.

3) Draw the energy diagram for the molecular orbitals of the N_2 molecule (include the atomic orbitals of each atom and the electrons provided by each atom, consider exclusively the valence shell). Label each type of orbital correctly. Calculate the bond order.

4) Draw all the molecular orbitals considered in Q3 (show clearly how they form through the overlap of the corresponding atomic orbitals). Indicate clearly orbital shape, phases, position of nuclei and nodes.

5) For the following molecules draw more than one possible Lewis structures, calculate the formal charges for each atom in each structure and choose the most representative structures for each case. Determine the geometry of each molecule: NF_3, ClO, ClO_2, H_2O_2, H_2CO, SO_2, CBr_4, PH_3, PBr_5.

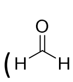
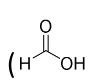
6) Indicate the hybridisation of the atoms shown in bold/underlined in the following molecules (for each case draw a representation of the hybrid atomic orbitals and also a representation of the molecule showing the bonding molecular orbital that is formed using the hybrid atomic orbitals).

   i) **<u>C</u>**H_4
   ii) **<u>O</u>**=C=O
   iii) O=**<u>C</u>**=O
   iv) **<u>N</u>**H_3

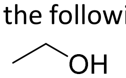
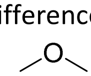
7) Are the molecules in Q5 polar or non-polar? Can they form hydrogen bonding-in which case-would they be donor or acceptors, or both? Discuss.

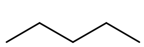
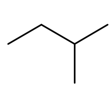
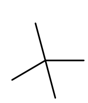
8) Predict if the following species will dissolve well in water or not and explain your choice: KBr, CH_4, CH_3OH, $CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH_2OH$, H_2O_2, CO_2, CCl_4, Br_2.

9) Explain the following trend in boiling point (°C):

   i) CH_4 : -161.5 CH_3F : -78.4 CH_3NH_2 : -6 CH_3OH: 64.7 CH_2O: -19 () CH_2O_2: 101 ()
   ii) CH_4 : -161.5 C_2H_6 : -89 C_3H_8 : -42 C_4H_{10} : 0 C_5H_{12} : 36 C_6H_{14} : 69 C_7H_{16} : 98 C_8H_{18} : 126 (all linear chains)

10) Explain the following differences in boiling point (°C) and explain the relationship between the molecules in each case:

   i)  78  -24

   ii)  36  28  9